

C Programming

BCA — Semester 1 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Explain the role of function in programming. How function is declared, defined and called in C? Write a program to find factorial of 'n' using recursive function.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is an Identifier and keyword? Explain the rules for naming valid identifiers in C with example. Write a C program to find sum of digits of a 'n' digit number.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How structure is different from union? Write a program to store and display basic information (roll, name, address, email, and phone) of students of using a structure.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is the basic structure of a C program? Explain each part.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. How break statement is different from continue statement. Explain with examples?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What do you mean by multi-dimensional array? Write program logic to add and display the sum of two m x n matrices.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Find errors (If any) in the following program and rewrite it.
`#define (studio.h)float main < >int m [ ] [ ] = {{1,2,3,4}};for (i = 4, i>=0 ; i--)for (j=0, j <4; j++);m [i] [j] = a[i][j] + b[i][j];return 0. 0;}`

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write a program to generate the following pattern of Integers.11 21 2 31 2 3 41 2 3 4 5

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Write a C program to check whether a given string is palindrome or not.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is pointer? Illustrate the use of a double pointer with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Why file is used in programming? Explain different file opening modes.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on:a. Macrosb. Null Pointer

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.